GAZIANTEP ISLAM SCIENCE AND TECHNOLOGY UNIVERSITY
FACULTY OF ENGINEERING AND NATURAL SCIENCES
INDUSTRIAL ENGINEERING DEPARTMENT
2024 EDUCATION PLAN

Semester 1						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE101	BASIC COMPUTER TECHNOLOGIES	Z	2	1	3	4
IE102	CALCULUS-I	Z	4	0	4	4
IE103	PHYSICS-I	Z	4	0	4	4
IE104	INTRODUCTION TO INDUSTRIAL ENGINEERING	Z	2	0	2	6
OZD021	ATATURK'S PRINCIPLES AND TURKISH REVOLUTION HISTORY-I	Z	2	0	2	2
OZD023	TURKISH-I	Z	2	0	2	2
OZD025	FOREIGN LANGUAGE-I	Z	2	0	2	2
OZD027	CAREER PLANNING	Z	1	0	1	2
EC-1	ELECTIVE COURSES	S	4	0	4	6
Toplam			**23**	**1**	**24**	**32**

EC-1 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE1XX	ENTREPRENEURSHIP	S	2	0	2	3
IE1XX	ENERGY EFFICIENCY AND APPLICATIONS	S	2	0	2	3

Semester 2						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE201	COMPUTER PROGRAMMING-I	Z	2	1	3	6
IE202	CALCULUS-II	Z	4	0	4	4
IE203	PHYSICS-II	Z	4	0	4	4
IE204	GENERAL CHEMISTRY	Z	3	0	3	6
OZD022	ATATURK'S PRINCIPLES AND TURKISH REVOLUTION HISTORY-I	Z	2	0	2	2
OZD024	TURKISH-I	Z	2	0	2	2
OZD026	FOREIGN LANGUAGE-I	Z	2	0	2	2
EC-2	ELECTIVE COURSES	S	4	0	4	6
Toplam			**23**	**1**	**24**	**32**

EC-2 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE2XX	DISASTER AND CRISIS MANAGEMENT	S	2	0	2	3
IE2XX	ENGINEERING ETHICS	S	2	0	2	3

Semester 3						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE301	OCCUPATIONAL HEALTH AND SAFETY- I	Z	2	0	2	3
IE302	COMPUTER PROGRAMMING-II	Z	3	0	3	4
IE303	SYSTEM ANALYSIS AND MODELING	Z	3	0	3	5
IE304	STATISTICS-I	Z	3	0	3	5
IE305	WORK AND TIME STUDY	Z	3	0	3	5
IE306	LINEAR ALGEBRA	Z	4	0	4	5
EC-3	ELECTIVE COURSES	S	4	0	4	6
Toplam			**22**	**0**	**22**	**33**

EC-3 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE3XX	INDUSTRY 4.0 AND DIGITAL TRANSFORMATION	S	2	0	2	3
IE3XX	INTRODUCTION TO INDUSTRIAL AND ORGANIZATIONAL PSYCHOLOGY	S	2	0	2	3

Semester 4						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE401	ERGONOMICS	Z	3	0	3	5
IE402	OCCUPATIONAL HEALTH AND SAFETY -II	Z	2	0	2	3
IE403	DIFFERENTIAL EQUATIONS	Z	4	0	4	5
IE404	STATISTICS- II	Z	3	0	3	5
IE405	NUMERICAL ANALYSIS	Z	3	0	3	5
IE406	TECHNICAL REPORT PREPARATION	Z	2	0	2	4
EC-4	ELECTIVE COURSES	S	4	0	4	6
Toplam			**21**	**0**	**21**	**33**

EC-4 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE4XX	CRITICAL THINKING METHODS	S	2	0	2	3
IE4XX	MARKETING	S	2	0	2	3

Semester 5						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE501	MANUFACTURING PROCESSES AND METHODS	Z	3	0	3	4
IE502	OPERATIONS RESEARCH- I	Z	4	0	4	6
IE503	PRODUCTION PLANNING- I	Z	3	0	3	5
IE504	MANAGEMENT INFORMATION SYSTEMS	Z	3	0	3	5
IE505	SIMULATION	Z	3	0	3	5
EC-5	ELECTIVE COURSES	S	5	0	5	8
Toplam			**21**	**0**	**21**	**33**

EC-5 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE5XX	HUMAN RESOURCE MANAGEMENT	S	3	0	3	5
IE5XX	MULTI-CRITERIA DECISION MAKING	S	3	0	3	5
IE5XX	PROGRAMMING LANGUAGES	S	3	0	3	5
IE5XX	FORECASTING METHODS	S	3	0	3	5
IE5XX	CURRENT TOPICS IN INDUSTRIAL ENGINEERING-I	S	3	0	3	5

Semester 6						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE601	PRODUCTION PLANNING- II	Z	3	0	3	5
IE602	FACILITIES PLANNING AND DESIGN	Z	3	0	3	5
IE603	STATISTICAL QUALITY CONTROL	Z	3	0	3	5
IE604	OPERATIONS RESEARCH- II	Z	4	0	4	6
IE605	TOTAL QUALITY MANAGEMENT	Z	3	0	3	4
EC-6	ELECTIVE COURSES	S	5	0	5	8
Toplam			**21**	**0**	**21**	**33**

EC-6 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE6XX	PROJECT MANAGEMENT	S	3	0	3	5
IE6XX	HEURISTIC OPTIMISATION	S	3	0	3	5
IE6XX	SUPPLY CHAIN, LOGISTICS AND WAREHOUSE PROCESSES MANAGEMENT	S	3	0	3	5
IE6XX	NETWORK THEORY	S	3	0	3	5
IE6XX	CURRENT TOPICS IN INDUSTRIAL ENGINEERING-II	S	3	0	3	5

Semester 7						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE701	ENGINEERING ECONOMY	Z	3	0	3	5
IE702	GRADUATION PROJECT-I	Z	0	4	2	10
EC-7	ELECTIVE COURSES	S	11	0	11	18
Toplam			**14**	**4**	**16**	**33**

EC-7 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE7XX	DECISION SUPPORT SYSTEMS	S	3	0	3	5
IE7XX	ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS	S	3	0	3	5
IE7XX	SPECIAL TOPICS IN OPERATIONS RESEARCH	S	3	0	3	5
IE7XX	FUZZY SET THEORY	S	3	0	3	5
IE7XX	DATABASE MANAGEMENT	S	3	0	3	5
IE7XX	DEEP LEARNING METHODS	S	3	0	3	5
IE7XX	STOCHASTIC PROCESSES AND APPLICATIONS	S	3	0	3	5
IE7XX	EXPERIMENTAL DESIGN AND ANALYSIS	S	3	0	3	5
IE7XX	ADVANCED MS EXCEL APPLICATIONS	S	3	0	3	5
IE7XX	TECHNOLOGY MANAGEMENT	S	3	0	3	5
IE7XX	INTRODUCTION TO DATA MINING	S	3	0	3	5

Semester 8						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
IE801	GRADUATION PROJECT-II	Z	0	4	2	5
IE802	APPLIED ENGINEERING TRAINING	Z	0	20	10	25
EC-8	ELECTIVE COURSES	S	2	0	2	3
Toplam			**2**	**24**	**14**	**33**

EC-8 Elective Courses						
Course Code	**Course Title**	**S/Z**	**T**	**U**	**Credit**	**ECTS**
OSDXX	UNIVERSITY COMMON ELECTIVE COURSE	S	2	0	2	3
IE8XX	ACADEMIC RESEARCH AND PRESENTATION TECHNIQUES	S	2	0	2	3

COURSE CONTENTS

SEMESTER 1 MANDATORY COURSES

BASIC COMPUTER TECHNOLOGIES (IE101)

Basic concepts related to information technologies, software, and hardware, generally operating systems, word processing programs, electronic spreadsheet programs, data presentation, internet usage in education, the impact of information technologies on social structure and their place in education, information systems security, and related ethical concepts.

Textbook:

- Applied and Illustrated Computer 1-2 (Basic Information Technologies), Zehra Alakoç Burma

Supplementary Textbook:

- Lecture notes

CALCULUS-I (IE102)

Functions, inverse functions, plotting graphs of simple curves, shifting graphs. Trigonometric functions, inverse trigonometric functions, logarithmic and exponential functions. Limit, rules of limit computation, continuity. Derivative of a function, geometric interpretation of the derivative, rules of differentiation, derivatives of trigonometric functions, inverse trigonometric functions, logarithmic and exponential functions. Higher-order derivatives, chain rule, derivative of an implicit function, applications of derivatives, and the concept of differentials. L'Hôpital's rule, concept of limit at infinity, Rolle's theorem and the Mean Value Theorem, extrema in functions. Asymptote concept, plotting graphs by studying the change in functions. Indefinite integrals. Methods of integral computation: change of variable, integration by parts, integration of polynomial, algebraic, and trigonometric (rational) functions. Riemann sums, definite integrals and their properties, fundamental theorem of calculus. Change of variable in definite integrals. Applications of definite integrals.

Textbook:

- Thomas, G.B., Finney, R.L. 2009; Calculus and Analytic Geometry, Volume I, Beta Publications, (Trans: Korkmaz, R.), Istanbul

Supplementary Textbook:

- Thomas, G.B., Finney, R.L. 2011; Calculus and Analytic Geometry, Volume II, Beta Publications, (Trans: Korkmaz, R.), Istanbul

- Balcı, M. 2009; General Mathematics 1, Balcı Publications, Ankara

PHYSICS-I (IE103)

This course covers topics such as kinematics in one and two dimensions, Newton's laws of motion, work and energy, conservation of energy, momentum, collisions, rotational dynamics, and static equilibrium. Through lectures, problem-solving sessions, and laboratory experiments, students will develop a conceptual understanding of physical laws governing motion and forces, and apply mathematical techniques to solve problems related to these concepts.

Textbook:

- "Fundamentals of Physics" by David Halliday, Robert Resnick, and Jearl Walker.

- "University Physics with Modern Physics" by Hugh D. Young and Roger A. Freedman.

Supplementary Textbook:

- Keller, F. J., 'Physics 1', trans. Ed. Akyüz R.Ö. and colleagues, Literatür Publishing, 2002, Istanbul

INTRODUCTION TO INDUSTRIAL ENGINEERING (IE104)

Basic Concepts, Topics, and Areas of Interest in Industrial Engineering, Engineering Ethics, Basic Statistics for Industrial Engineering, Engineering Economy, Operations Research, Work and Method Study with Ergonomics, Production Planning and Control with Material and Inventory Management, Quality Management, Facility Management, Production and Service Systems, Simulation.

Textbook:

- "Introduction to Industrial and System Engineering", Kula, U., Torkul, O., Taşkın, H., Değişim Publications, 2006

Supplementary Textbook:

- "Introduction to Industrial Engineering", Editor: Öztemel, E., Papatya Publishing, 2009

- "Introduction to Industrial and System Engineering", Turner, W.C., Mize, J.H., Case, K.E., Nazemitz, J.W., Pearson, 1993

ATATURK'S PRINCIPLES AND TURKISH REVOLUTION HISTORY-I (OZD021)

Events, ideas and principles in the process of the birth and development of modern Turkey

Textbook:

- Kemal Atatürk, Nutuk I, II, III, Istanbul, 1967. Atatürk's Principles and History of Revolution I, II, III, Higher Education Council Presidency Publications, Ankara, 1990.

Supplementary Textbook:

- Mehmet Alpargu-İsmail Özçelik-Nuri Yavuz, Atatürk's Principles and History of Revolution, Ankara, 2003.

- Atatürk article, Islamic Encyclopaedia, Ministry of National Education, Ankara, 1990

- Atatürk's Revolution, Ministry of Culture, Ankara, 1990

TURKISH-I (OZD023)

Turkish language and other languages, phonetics, words in terms of meaning and functions, semantics, sentence knowledge.

Textbook:

- Kavaz, İ. (Ed.) 2009; Turkish Language and Composition Information, Data Publications, Ankara.

Supplementary Textbook:

- Ergin, M. 1995; Turkish Language for Universities, Bayrak Publications, Istanbul. Banguoğlu, T. 1974;Grammar of Turkish, Baha Matbaası, Istanbul.

FOREIGN LANGUAGE-I (OZD025)

This course is designed to consolidate the student's working knowledge of the English language through reinforcement of reading comprehension, listening and writing skills in academic English.

CAREER PLANNING (OZD027)

Career planning is a strategic process that guides individuals through the exploration, development, and implementation of their professional paths. This course aims to equip students with the knowledge, tools, and skills necessary to navigate their career journeys effectively. Through a combination of lectures, workshops, and practical assignments, students will learn to assess their interests, values, and skills; explore career options and market trends; set realistic career goals; and develop actionable plans to achieve those goals. Topics include self-assessment techniques, career exploration strategies, networking and informational interviewing, resume and cover letter writing, interview preparation, and strategies for ongoing professional development.

Textbooks:

- Career Development and Planning: A Comprehensive Approach" by Robert C. Reardon, Janet G. Lenz, James P. Sampson, and Gary W. Peterson

- Do What You Are: Discover the Perfect Career for You Through the Secrets of Personality Type" by Paul D. Tieger, Barbara Barron, and Kelly Tieger.

SEMESTER 1 ELECTIVE COURSES

ENTREPRENEURSHIP (IE1XX)

Entrepreneurship courses are designed to inspire and equip students with the knowledge and skills necessary to conceive, launch, and grow new business ventures. The curriculum typically explores the fundamentals of business creation and development, including idea generation, market research, business model development, funding strategies, and the management of growth and innovation. Emphasis is placed on critical thinking, problem-solving, and the entrepreneurial mindset, encouraging students to develop resilience, creativity, and the ability to navigate the complexities of starting and running a business. The course also incorporates case studies of successful entrepreneurs, hands-on projects, and pitch presentations to provide practical experience in the entrepreneurial process.

Textbook:

- "Entrepreneurship: Successfully Launching New Ventures" by Bruce R. Barringer and R. Duane Ireland.

- "The Lean Startup: How Today's Entrepreneurs Use

ENERGY EFFICIENCY AND APPLICATIONS (IE1XX)

The aim of the course is to give the student basic information about energy management, energy efficiency and alternative energy sources and applications. Within the scope of this course; The importance of energy saving, the general energy situation / structure of Turkey and the Turkish Industry, the production, consumption and management of primary energy resources, energy efficiency and increasing applications, environment-energy relationship, applicability and analysis of renewable energy resources, applications that increase energy efficiency in different sectors are included.

SEMESTER 2 MANDATORY COURSES

COMPUTER PROGRAMMING-I (IE201)

For a course on Computer Programming with Python, the content could focus on introducing students to the fundamentals of programming using Python, a versatile and widely-used programming language. The course would start with Python syntax and basic programming concepts such as variables, data types, and control structures. Students will learn to write their own Python scripts to solve problems, manage data, and create simple applications. The course will also cover more advanced topics such as object-oriented programming, error handling, and working with external libraries to extend Python's capabilities. Practical assignments and projects will be integrated throughout the course to reinforce learning and develop problem-solving skills in real-world scenarios.

Textbook:

- "Python Crash Course" by Eric Matthes

- "Automate the Boring Stuff with Python" by Al Sweigart

CALCULUS- II (IE202)

Matrices, determinants, eigenvalues and eigenvectors, inverse matrices. Solution of linear equation systems and Cramer's method using echelon form. Conic sections and quadratic equations, polar coordinates and graph plotting, parametrization of curves in the plane. Three-dimensional space and Cartesian coordinates. Vectors in the plane and space. Dot, cross, and mixed products. Lines and planes in three-dimensional space. Cylinders, cones, and spheres. Cylindrical and spherical coordinates. Vector-valued functions and curves, curvature, torsion, and TNB frame in space. Multivariable functions, limit, continuity, and partial derivatives. Chain rule, directional derivatives, Gradient, Divergence, Curl, and tangent planes. Extremum values and saddle points, Lagrange multipliers, Taylor and Maclaurin series. Double integrals, area, moment, and center of mass. Double integrals in polar form. Triple integrals in Cartesian coordinates. Mass, moment, and center of mass in three-dimensional space. Triple integrals in cylindrical and spherical coordinates. Change of variables in multiple integrals. Line integrals, vector fields, work, flux. Green's Theorem in the plane. Surface area and surface integrals. Stokes' Theorem, Divergence Theorem, and applications.

Textbook

- Thomas, G.B., Finney, R.L. 2009; Calculus and Analytic Geometry, Volume I, Beta Publications, (Translated by: Korkmaz, R.), Istanbul.

Supplementary Textbook:

- Thomas, G.B., Finney, R.L. 2011; Calculus and Analytic Geometry, Volume II, Beta Publications, (Translated by: Korkmaz, R.), Istanbul.

- Balcı, M. 2009; General Mathematics 1, Balcı Publications, Ankara.

PHYSICS-II (IE203)

Coulomb's Law, Electric Field, Electric Flux, Gauss's Law, Electrical Potential, Capacitors, Formation of Current and Resistance, Direct Current Circuits, Kirchhoff's Laws, Magnetic Field, Biot-Savart Law, Ampere's Law, Induction, Faraday's Law, Lenz's Law, Inductance, Energy in Magnetic Field, Oscillations in LC Circuits, Electromagnetic Waves.

Textbook:

- Physics II for Science and Engineering (Electricity and Magnetism), R.A.Serway; Translation Editor: Kemal Çolakoğlu, (Translated from the 5th edition), Palme Publishing, 2002.

- Physics II (Electricity), F.J.Keller, W.E.Gettys, M.J.Skove, Translation Editor: R.Ömür Akyüz, Literatür Publishing, 2006.

GENERAL CHEMISTRY (IE204)

Properties and Measurement of Matter. Atoms and Atomic Models. Periodic Table. Concept of Mole. Calculations and Chemical Compounds. Calculation of Chemical Formulas. Naming Compounds. Chemical Reactions and Calculations. Aqueous Solution Reactions. Oxidation-Reduction Reactions and Redox. Solutions and Gases. Gas Laws and Manometers. Gas Mixtures and Diffusion Rates of Gases. Examination of the Periodic Table, Formation of Chemical Bonds, Organic Chemistry, Intermolecular Forces. Chemical Kinetics and Equilibrium, Redox Reactions and Electrochemistry.

Textbook:

- Raymond Chang, General Chemistry Basic Concepts, Editors: Tahsin Uyar, Serpil Aksoy, Recai İnam, Palme Publishing, Ankara, 2021.

Supplementary Textbook:

- Petrucci, Herring, Madura, Bissonnette, General Chemistry Principles and Modern Applications, Palme Publishing, 2012.

ATATURK'S PRINCIPLES AND TURKISH REVOLUTION HISTORY-II (OZD022)

Declaration of the Republic and its political, social, cultural, and economic reforms; Issues unresolved by the Treaty of Lausanne and their resolutions; Movements against the new order; Attempts at multi-party life and their outcomes; Turkish Foreign Policy during the Atatürk era, İnönü era, and the years of World War II; Turkey's transition to multi-party politics, the era of the Democratic Party, and thereafter; Atatürkism and Atatürk's Principles.

Textbook:

- Mumcu, A., Özbudun, E., Feyzioğlu, T., Ülken, Y., Çubukçu, A. 1992. Principles of Atatürk and History of Reforms, Higher Education Council Publications, Ankara Atatürk, M. K., 2005.

- Nutuk (The Speech), Alfa Publications, Istanbul Mehmet Alpargu-İsmail Özçelik-Nuri Yavuz, Principles of Atatürk and History of Reforms, Ankara, 2003 Atatürk article, İslam Encyclopedia, Ministry of National Education, Ankara, 1990.

- Atatürk's Revolution, Ministry of Culture, Ankara, 1990.

TURKISH-II (OZD024)

Written Expression, Method and Plan in Written Expression, Written Expression Application, Scientific Texts (Article-Report-Critique), Official Texts.

Textbook:

- Instructor's notes

- Turkish Language and Composition Information, Commission, Judiciary Publishing House, Ankara, 1984.

FOREIGN LANGUAGE-II (OZD026)

Academic reading comprehension skills are further reinforced through and intensive reading of both scientific and academic texts and tecniques of writing are further developed with emphasis on the usage of technical vocabulary.

SEMESTER 2 ELECTIVE COURSES

DISASTER AND CRISIS MANAGEMENT (IE2XX)

The Disaster and Crisis Management course is designed to equip students with the knowledge and skills necessary to effectively manage and respond to various types of disasters and crises, including natural, technological, and human-caused events. The curriculum covers the principles of emergency management, disaster preparedness, response strategies, recovery processes, and mitigation efforts. Students will learn about the roles and responsibilities of different stakeholders, including government agencies, non-profit organizations, and community members, in disaster management. The course also emphasizes the importance of communication, coordination, and collaboration among various entities during a crisis. Through case studies and simulations, students will gain practical experience in developing disaster response plans, conducting risk assessments, and implementing resilience strategies to minimize the impact of future disasters.

Textbook:

- "Introduction to Emergency Management" by George Haddow, Jane Bullock, and Damon P. Coppola.

- "Disaster Policy and Politics: Emergency Management and Homeland Security" by Richard Sylves.

ENGINEERING ETHICS (IE2XX)

The aim of this course is; The importance of the individual and universal freedom dimension of ethics, understanding universal ethical rules and values, learning the principles of engineering ethics and professional responsibility, and providing students with the ability to question professional dilemmas based on these.

Textbook:

-“Engineering Ethics: Concepts and Cases” by Jr. Charles E. Harris, M. S. Pritchard, M.J. Rabins, R. James, E. Englehardt, 6th Edition, Cengage Learning

SEMESTER 3 MANDATORY COURSES

OCCUPATIONAL HEALTH AND SAFETY- I (IE301)

Work accidents, occupational diseases, laws and social order rules, personal protective equipment, ergonomics, fire and fire protection, first aid.

Textbook

- Occupational Safety CASGEM notes, Occupational Health and Safety, Chamber of Civil Engineers notes.

COMPUTER PROGRAMMING-II (IE302)

This Object-Oriented Programming (OOP) with Python course will introduce you to the principles of OOP, a programming paradigm that models complex problems as interactive objects. You will learn about classes, instances, inheritance, encapsulation, polymorphism, and abstraction. The course covers the creation and manipulation of objects, leveraging Python's simplicity and flexibility to build robust and scalable applications. By understanding OOP concepts, students will be able to design software that is easy to maintain, extend, and scale.

Textbook:

- "Python 3 Object-Oriented Programming" by Dusty Phillips

- "Learning Python" by Mark Lutz

SYSTEM ANALYSIS AND MODELLING (IE303)

Change in System Paradigms, General System Theory, System Approach, System Analysis, System Design, Systemic Problems, Modeling of Systems, System Analysis and Design Process, Basic Stages of System Analysis Work, System Engineering, Structure and Behavior of Dynamic Systems, System Life Cycle, Methods and Tools Used in System Analysis Work, Examples of System Analysis Applications.

Textbook

- Whitten, Bentley, Barlow, System Analysis & Design Methods, IRWIN, Second Edition, 1992.

Supplementary Textbook:

- John Sterman, Business Dynamics: Systems Thinking and Modeling for a Complex World, McGraw-Hill/Irwin, 2000.

STATISTICS- I (IE304)

Data Organization and Analysis, Introduction to Probability, Probability Rules, Random Variables.

Textbook:

- Akdeniz, Fikri., Probability and Statistics, 13th Edition.

Supplementary Textbook:

- Lecture Notes

WORK AND TIME STUDY (IE305)

Basic concepts related to work study, method study, time study, work sampling, group timing technique, synthetic timing techniques, and MTM applications.

Textbook:

- Modelling for Management: Simulation in Support of Systems Thinking by George P. Richardson

LINEAR ALGEBRA (IE306)

Linear equation systems solution methods (Cramer's rule, inverse matrix, methods for reducing to normal form), matrix and determinant operations, eigenvalues and eigenvectors of a matrix, linear transformations in linear spaces.

Textbook:

- Gözükızıl, Ö. Faruk, Linear Algebra, Değişim Publications, Istanbul, 2000

Supplementary Textbook:

- Lipschutz, S., Hacısalihoğlu, H., Akın, Ö., Linear Algebra: Theory and Problems, Nobel Publishing Distribution, Ankara, 1991.

- Yusubov, İ., Panahov, M., Elements of Linear Algebra and Finite Dimensional Linear Operator Theory, Sakarya Publishing House, Sakarya, 2006.

SEMESTER 3 ELECTIVE COURSES

INDUSTRY 4.0 AND DIGITAL TRANSFORMATION (IE3XX)

The course on aims to equip students with an in-depth understanding of the fourth industrial revolution and its impact on businesses and society. It covers the foundational concepts of smart manufacturing, including the Internet of Things (IoT), big data analytics, artificial intelligence (AI), and cyber-physical systems. Students will explore how these technologies drive efficiency, sustainability, and innovation across various sectors, transforming traditional production and management processes into more agile, data-driven operations. The course also examines the ethical, socio-economic, and security challenges posed by digital transformation, preparing students to navigate and lead in the evolving industrial landscape. Through case studies, interactive projects, and critical analysis, learners will gain practical insights into implementing Industry 4.0 strategies and fostering digital culture within organizations.

Textbook:

- "The Fourth Industrial Revolution" by Klaus Schwab

- "Digital Transformation: Survive and Thrive in an Era of Mass Extinction" by Thomas M. Siebel

INTRODUCTION TO INDUSTRIAL AND ORGANIZATIONAL PSYCHOLOGY (IE3XX)

The aim of this course is to interpret the basic concepts and theories in the field of Industrial and Organizational Psychology and to have an idea about how psychological science is integrated into business life with examples from business life and theoretical knowledge. Students will be able to

explain the problems experienced in the workplace with the theories they have learned and to find solutions to the problems encountered in business life (individual-group-organization) within the framework of the knowledge they have acquired. They will be able to explain about Human Resources functions, which are the basic subjects of Industrial Psychology, and the main subjects of Organizational Psychology.

Texbook:

- “Introduction to Industrial and Organizational Psychology” Ronald E. Riggio,

SEMESTER 4 MANDATORY COURSES

ERGONOMICS (IE401)

The course will cover a wide range of topics, including the ergonomic design of workspaces, tools, and equipment to fit the user's physical capabilities and limitations; the study of workplace ergonomics to prevent musculoskeletal disorders; cognitive ergonomics focused on mental processes, such as perception, memory, reasoning, and motor response, as they affect interactions among humans and other elements of a system; and the application of ergonomic principles in varied contexts, from office environments to industrial settings. Students will engage in practical exercises to assess work environments, design ergonomic solutions, and evaluate their effectiveness in improving comfort, safety, and productivity.

Textbook:

- "Ergonomics: How to Design for Ease and Efficiency" by K.H.E. Kroemer, H.B. Kroemer, and K.E. Kroemer-Elbert.

OCCUPATIONAL HEALTH AND SAFETY-II (IE402)

Industrial accidents and their causes - The importance of workplace planning based on potential industrial hazards - Development of engineering and managerial methods to control industrial hazards - Evaluation of the effectiveness of the measures to be taken

Textbook:

- CASGEM notes on occupational safety

- Occupational health and safety, IMO notes

- Occupational Safety, Ercüment DİZDAR, ABP Publishing House, 2010.

DIFFERENTIAL EQUATIONS (IE403)

Basic concepts and classification of differential equations, First order equations and engineering applications, Second and higher order differential equations and engineering applications, Equations with variable coefficients, Systems of linear equations: Scalar and matrix methods, Laplace transform, Engineering applications, Introduction to numerical solution of differential equations

Textbook:

- Çengel, Y. A. and Palm, W. J. (Turkish: Tahsin Engin), 2012, Differential Equations for Engineers and Scientists, Güven Kitabevi, Izmir.

STATISTICS-II (IE404)

Statistical distributions, Point and interval estimates, one sample hypothesis tests, two sample hypothesis tests

Textbook:

- Akdeniz, F., Probability and Statistics, 13th Edition

NUMERICAL ANALYSIS (IE405)

The course provides an approach to understanding mathematical methods for solving engineering problems, the algorithms used and their performance. The course covers root finding, basic linear algebra, solving systems of linear equations, curve fitting, solutions of simple differential equations and unconstrained optimization. The concept of algorithms and applications of iterative algorithms, which are frequently encountered in industrial engineering, are introduced.

Textbook:

- Ibrahim Uzun, Numerical Analysis, 6th Edition

TECHNICAL REPORT PREPARATION (IE406)

The concept of knowledge and science, Aims and types of scientific research, research methods, planning of research, data collection and data analysis, survey technique, transformation of scientific research results into a report.

Textbook:

- Rauf Arıkan; Research Techniques and Report Writing, Gazi Kitabevi

- Hamza AL; Scientific Research Methods, Sakarya Publishing

SEMESTER 4 ELECTIVE COURSES

CRITICAL THINKING METHODS (IE4XX)

The aim of this course is; To provide students with the ability to think critically and develop solutions to life events, professional problems and situations before graduation.

Textbook:

- “Critical Systems Thinking: A Practitioner’s Guide” by Michael C. Jackson, 1st Edition, Wiley

MARKETING (IE4XX)

The aim of this course is; To explain the basics of marketing management, one of the basic functions of companies, and today's modern marketing methods.

Textbook:

- “Principles of Marketing” by Gary Armstrong

SEMESTER 5 MANDATORY COURSES

MANUFACTURING PROCESSES AND METHODS (IE501)

Main concepts of production, product manufacturing relationship, casting and casting methods. Cold and hot metal shaping. Powder metallurgy. Turning, milling, drilling, shaping, grinding and sawing production methods. Chip removal theory and machining methods.

Textbook:

- Manufacturing Processes for Industrial Engineers, D. EROL and B. ULUTAŞ, Seçkin Publishing House, 2011

- Fundamentals of Modern manufacturing: Materials, Processes, and Systems, 2nd edition, M. P. Groover, John Wiley & Sons Inc., 2002.

OPERATIONS RESEARCH-I (IE502)

Linear decision models. Fundamentals of linear programming and solution approaches. Solution approaches of linear decision model. Sequential numerical solution of linear decision model. Simplex algorithm. Use of artificial variables.

Textbook:

- Aladağ, Z., 2014.Operations Research-I, Umuttepe Publications, Kocaeli University, Faculty of Engineering. Department of Industrial Engineering

Supplementary Textbook:

- Kara, I. 2000. Linear Programming, Bilim Teknik Publishing House, Başkent Univ., Faculty of Engineering, Department of Industrial Engineering, Ankara.

PRODUCTION PLANNING-I (IE503)

Introduction, Historical Development of Production Systems, Demand Forecasting, Material Management, ABC Analysis, Multi-Criteria ABC Analysis, Aggregate Production Planning - Table Method, Aggregate Production Planning - Transportation Model, Aggregate Production Planning - Linear Programming, Master Production Scheduling, Inventory management, Material Requirements Planning (MRP), Lot Size Determination, Capacity Planning

Textbook:

- The Fundamentals of Production Planning and Control, Pearson, 2006. (S.N.Chapman) Production&Operations Analysis (S. Nahmias) Üretim Planlama ve Kontrol, Murat Baskak, Prof. Dr. Mehmet Tanyaş, İRFAN YAYINEVİ

Supplementary Textbook:

- Production Organization and Management, Mete Doğruer, 2005

- Operation Management Quality and Competitiveness in a Global Environment, Roberta S. Russell Bernard W. Taylor III, 2006Yardımcı Ders Kitabı

MANAGEMENT INFORMATION SYSTEMS (IE504)

Introduction to management information systems, Electronic data movement systems, Office automation systems, Management reporting systems, Artificial Intelligence Supported information systems.

Textbook:

- Management Information Systems: Managing Digital Business, J.P. Laudon, K.C. Laudon, Nobel Publishing Distribution

Supplementary Textbook:

- Information Systems: A Management Perspective, S. Alter, Addison-Wesley

SIMULATION (IE505)

Introduction to Simulation and Basic Concepts, Simulation model structure and types, Random Number and Variable generation and testing, Data analysis and conformance testing, Simulation in Queuing and Service systems, Simulation in Inventory Management, Maintenance and replacement management models, Simulation Modeling in Manufacturing systems, Simulation packages.

Textbook:

- Simulation modeling and analysis by Averill M. Law, W. David Kelton

Supplementary Textbook:

- Modelling for Management: Simulation in Support of Systems Thinking by George P. Richardson

- Simulation Modeling (The Wiley series in management and administration) by Forest Paul Wyman

SEMESTER 5 ELECTIVE COURSES

HUMAN RESOURCE MANAGEMENT (IE5XX)

Introduction to human resource management, Equal opportunity and laws, Job analysis and talent management process, Personnel planning and recruitment, Worker testing and selection, Training and development, Performance management and evaluation, Establishing strategic pay plans, Performance and incentive based pay systems, Labor relations and collective bargaining, Safety health and risk management, Global human resource management.

Textbook:

- "Fundamentals of Human Resource Managament" by Dessler, G., 2013, Pearson

MULTI-CRITERIA DECISION MAKING (IE5XX)

Basic concepts in decision making theory, modelling and solution methods of decision making problems with applications, decision making methodology.

Textbook:

-"Multi-Criteria Decision Making Methods: A Comparative Study" by Evangelos Triantaphyllou, 2000, Springer

PROGRAMMING LANGUAGES (IE5XX)

Design and implementation principles of programming languages, language syntax (semantics, BNF, allocation), language operators (compilers, interpreters), representations (data structures, control structures and bindings) and styles (procedural, functional programming, logic programming, modular programming, object oriented programming), term project.

Textbook:

- "Concepts of Programming Languages" by Sebestra R.W., 10th Ed., PEARSON

- "Programming Languages Design and Implementation" by Pratt T.W., Zelkowitz M.V., Prentice Hall

FORECASTING METHODS (IE5XX)

The aim of the course is to have knowledge about forecasting, which is a sub-element of decision making about the future, to ensure that the correct forecasting technique is applied for the situation encountered, to gain the ability to use the necessary statistical and computer-aided programs for forecasting and to interpret the results correctly.

Textbook:

- Hanke, J.E. and D.W. Wichern (2008). Business Forecasting. 8thEdition, Pearson Education International; Harlow, Essex.

CURRENT TOPICS IN INDUSTRIAL ENGINEERING-I (IE5XX)

The aim of this course is to explain the effects and results of technological and industrial developments on industrial engineering.

SEMESTER 6 MANDATORY COURSES

PRODUCTION PLANNING-II (IE601)

Lot Size Determination Methods in Material Requirements Planning Scheduling Assembly Line Balancing Manufacturing Resource Planning (MRP II), Enterprise Resource Planning (ERP) Supply Chain Management and Logistics Just in Time Production Systems

Textbook:

- The Fundamentals of Production Planning and Control, Pearson, 2006. (S.N.Chapman)

- Production&Operations Analysis (S. Nahmias) Üretim Planlama ve Kontrol, Murat Baskak, Prof. Dr. Mehmet Tanyaş, İRFAN YAYINEVİ

FACILITIES PLANNING AND DESIGN (IE602)

General definition and basic concepts, Systematic Plant Layout (SLP) and Production Flow Analysis (PFA), Determination of Flow, Area and Activity relationships and drawing diagrams, Material handling systems and tools, Layout planning models and design algorithms, Computer aided plant design, Factory and office layout, Plant Location Selection models and techniques, Assembly line balancing, Plant planning project presentation

Textbook:

- Facility Design from Strategy to Practice, ERKUT, H., BASKAK, M., İrfan Publishing, 1997, ISBN: 975-371-035-6

Supplementary Textbook:

- Facilities Planning 3E WIE James A. TOMPKINS, JOHN A. WHITE, Yavuz A. Bozer, J. M. A. Tanchoco John Wiley Higher Education , 2003 ISBN: 0471389374

- Facilities and Workplace Design, An Illustrated Guide, Quarterman Lee, Arild Amundsen, William Nelson and Herbert Tuttle, Strategos 2007.

STATISTICAL QUALITY CONTROL (IE603)

Quality and general quality definitions, quality systems, probability distributions, statistical process control, control chart structure and types, acceptance sampling and types, design of experiments, Taguchi technique

Textbook:

- Introduction Statistical Quality Control, third ed. D.C.Montgomery, John Wiley & Sons, 1997.

Supplementary Textbook:

- Principles of Quality Control, Jerry Banks, John Wiley & Sons, 1989.

OPERATIONS RESEARCH-II (IE604)

Integer Programming, Goal Programming, Network Models, Markov Chains, Game Theory, Dynamic Programming, Introduction to Nonlinear Programming

Textbook:

- Operations Research, T. L. Winston, PWS Publishing Company, 1997.

TOTAL QUALITY MANAGEMENT (IE605)

Total Quality Management (TQM) is a comprehensive and structured approach to organizational management that seeks to improve the quality of products and services through ongoing refinements in response to continuous feedback. The course content will cover the history and evolution of quality management, principles and practices of TQM, including the key concepts of customer satisfaction, employee involvement, continuous process improvement, and performance measurement. It will also delve into the tools and techniques for quality control and improvement, such as Six Sigma, Lean Manufacturing, and ISO 9000 standards, along with case studies from various industries to illustrate the practical application of TQM principles. The course aims to equip students with the knowledge and skills to implement TQM practices effectively in their organizations to enhance competitiveness and customer satisfaction.

Textbook:

- Total Quality Management: Text, Cases, and Readings" by Joel E. Ross and Susan Perry.

- Quality Management for Organizational Excellence: Introduction to Total Quality" by David L. Goetsch and Stanley Davis

SEMESTER 6 ELECTIVE COURSES

PROJECT MANAGEMENT (IE6XX)

Introduction to Project Management, Project Management Process, Introduction to Project Planning, CPM, PERT, Project Resource Management, Project Performance Measurement, Project Risk Management, Project Quality and Human Resources Management, MS Project Application, Case Studies

Textbook:

- Burke Rory, Project Management Planning and Control, John Wiley & Sons Inc, 1996.

Supplementary Textbook:

- Chapman C., and Ward S., "Project Risk Management: Processes, Techniques and Insights", 2nd Edition, ISBN: 978-0-470-85355-9, September 2003, ©2003

- Kazan H., Project management, Istanbul University Faculty of Open and Distance Education

HEURISTIC OPTIMIZATION (IE6XX)

Although most discrete optimization research relies on the exact solution of integer or combinatorial problems, there are numerous applications that use some form of heuristics. Heuristic methods are methods that look for a good solution in a reasonable amount of time, but not necessarily an optimal solution. The purpose of this course is to examine a wide range of heuristics, explaining to students their general properties and limitations, and the types of problems to which they are best adapted.

Textbook:

-"Heuristics for Optimization and Learning" by F. Yalaoui, L. Amodeo, E-G. Talbi, 7th Edition, Springer

SUPPLY CHAIN, LOGISTICS, AND WAREHOUSE PROCESSES MANAGEMENT (IE6XX)

The aim of this course is to emphasize the importance of efficient and effective management of supply chain, logistics and warehouse processes in the fierce competitive environment of today and the future. This course content includes; Basic concepts related to supply chain management and logistics, supply chain analysis, the place and importance of logistics and warehouse processes in the supply chain, basic warehouse processes, and approaches applied for effective management of warehouse management processes will be discussed.

Textbook:

-"Logistics and Supply Chain Management" by Martin Christopher, 6th Edition, Pearson

NETWORK THEORY (IE6XX)

Introduction to graph theory; shortest path and related algorithms; network flow algorithms; matching and covering algorithms; traveling salesman problem and extensions; Chinese postman problem and extensions; location problems on a network; stochastic network.

Textbook:

- Operations Research, T. L. Winston, PWS Publishing Company, 1997.

CURRENT TOPICS IN INDUSTRIAL ENGINEERING (IE6XX)

The aim of this course is to explain the effects and results of technological and industrial developments on industrial engineering.

SEMESTER 7 MANDATORY COURSES

ENGINEERING ECONOMY (IE701)

Basic Concepts, Balancing Cash Flows, Interest and the Time Value of Money, Present Value Method, Annual Equivalent Method, Future Value Method, Internal Rate of Return Method, External Rate of Return Method, Evaluation of Alternatives, Depreciation, Post-Tax Cash Flow Analyses, The Effects of Inflation on Cash Flows, Break-Even Analysis, Cost-Benefit Analysis, Replacement Analyses

Textbook

- "Engineering Economy", Degarmo, E.P., Sullivan, W.G., Bontadelli, J.A., 2008

- "Engineering Economy", Okka, O., 2010

GRADUATION PROJECT-I (IE702)

Each student will carry out a project study on topics related to the industrial engineering department, present it in a report form to their advisor faculty member, and defend it.

SEMESTER 7 ELECTIVE COURSES

DECISION SUPPORT SYSTEMS (IE7XX)

Definition, Components, and Benefits of Decision Support Systems, Classification of Decision Support Systems, Decisions and Decision Makers, Modelling the Decision Making Process, Data Warehouses and Data Mining, Business Intelligence, Interactive and Computer-Assisted Technologies, Group Decision Support Systems, Information Management Systems, Applications of Decision Support Systems

Textbook

- Marakas, G.M., Decision Support Systems, Prentice-Hall, 2003.

- Turban, E., Aronson, J.E., Liang, T.P., Decision Support Systems and Intelligent Systems, Prentice-Hall, 2005.

- Laudon, K.C., Laudon, J.P., Management Information Systems, 9th Ed., Prentice-Hall, 2005.

ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS (IE7XX)

Introduction to Artificial Intelligence and Expert Systems, Supervised and Unsupervised Learning, Collective Learning (Ensemble Learning), Modelling with Unsupervised Learning, Artificial Neural Networks, Natural Language Processing, Reinforcement Learning, Deep Learning, Fuzzy Logic and Fuzzy Set Theory, Introduction to Heuristic Search Techniques, Python Applications

Textbook:

- Joshi, P., Artificial Intelligence with Python, Packt, Birmingham-Mumbai, ISBN 978-1-78646-439-2

- Das, P.K., Rao, D.C., Sahu, K.K., Artificial Intelligence Lecture Notes, for Bachelor of Technology in Computer Science and Engineering & Information Technology

- Sarkar, D., Bali, R., Sharma, T., Practical Machine Learning with Python, A Problem-Solver's Guide to Building Real-World Intelligent Systems

SPECIAL TOPICS IN OPERATIONS RESEARCH (IE7XX)

The course "Special Topics in Operations Research" is designed to delve into advanced and emerging areas within the field of operations research, focusing on the development and application of analytical methods to help make better decisions. Topics may include, but are not limited to, stochastic models, simulation, optimization under uncertainty, game theory, queueing theory, and network models. Students will explore the latest research and methodologies in operations research, applying these concepts to complex problems in various industries such as logistics, healthcare, finance, and manufacturing. The course will also cover the use of software tools and programming languages relevant to operations research, enhancing students' ability to model, analyse, and solve real-world problems.

Textbooks:

- Operations Research: Applications and Algorithms" by Wayne L. Winston.

- Introduction to Operations Research" by Frederick S. Hillier and Gerald J. Lieberman

FUZZY SET THEORY (IE7XX)

Fuzzy Set Theory course delves into the principles and mathematical structure of fuzzy sets, a concept introduced by Lotfi A. Zadeh in 1965 as an extension of classical set theory. It explores the theory's foundational elements, including basic definitions, operations on fuzzy sets, and properties. The course also covers advanced topics such as fuzzy numbers, fuzzy relations, fuzzy logic, and their applications in decision-making, control systems, and information processing. Throughout, the course emphasizes the practical implications of fuzzy set theory in handling uncertainty and imprecision in various scientific and engineering domains. This theoretical framework provides a robust tool for modelling complex systems where classical logic and set theory fall short.

Textbooks:

- Fuzzy Sets and Fuzzy Logic: Theory and Applications" by George J. Klir and Bo Yuan

- Fuzzy Set Theory - and Its Applications" by Hans-Jürgen Zimmermann

DATABASE MANAGEMENT (IE7XX)

The aim of this course is to provide students with knowledge and experience about the importance, requirements, design and development of database management systems. To explain the basic definitions and concepts in this subject, to design a database using different database management system (DBMS) tool software, to define the data, to store and effectively manage interrelated data groups in the computer environment, data modeling, logical design and concepts, and SQL, which allows them to manage the data. Teaching the query language and developing database applications with Microsoft Access, Microsoft SQL Server or MySQL DBMS, which are widely used relational database management systems, are the basic content of this course.

Textbook:

-"Database Management Systems" by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition, McGraw-Hill

DEEP LEARNING METHODS (IE7XX)

The purpose of this course is to introduce the topic of deep learning. Deep learning is a branch of machine learning that deals with the development and implementation of modern neural networks. This course content includes; Applications in fields such as basic neural networks, convolutional and recursive network structures, image processing and natural language processing will be explained.

Textbook:

-"Computational Methods for Deep Learning: Theory, Algorithms, and Implementations" by Wei Qi Yan, 2nd Edition, Springer

STOCHASTIC PROCESSES AND APPLICATIONS (IE7XX)

This course on Stochastic Processes and Applications provides a comprehensive overview of the theory and practical applications of stochastic processes. Students will explore the foundational concepts of probability theory, delve into the properties and behaviours of discrete and continuous-time stochastic processes, including Markov chains, Poisson processes, and Brownian motion. The course emphasizes the mathematical underpinnings of stochastic processes, as well as their applications in various fields such as finance, insurance, engineering, and the natural sciences. Through lectures, problem-solving sessions, and projects, students will learn to model real-world phenomena, analyse data, and develop a deep understanding of the randomness and uncertainty inherent in complex systems.

Textbook:

- "Introduction to Stochastic Processes" by Gregory F. Lawler

- "Stochastic Processes: Theory for Applications" by Robert G. Gallager

EXPERIMENTAL DESIGN AND ANALYSIS (IE7XX)

The aim of this course is; to introduce experimental design concepts and statistical modeling tools related to data analysis. Students are expected to learn how to plan, design, and conduct experiments efficiently and effectively, as well as how to analyze collected data to make recommendations to decision makers. . The tools and techniques covered can easily be used to solve real-life problems in new product design and development, process development, and manufacturing process improvement. Computer software packages (Design Expert, Minitab) can be used to demonstrate the methods presented throughout the course.

Textbook:

-"Desing and Analysis of Experiments" by Douglas C. Montgomery, 8th Edition, Wiley

ADVANCED MS EXCEL APPLICATIONS (IE7XX)

Excel Power Query, Scenario Manager, Forecasting, Pivot Tables, Advanced Formulas, Macros, Power Pivot, basic features of programming, variable, constant, command, procedure concepts, loop, teaching commands, writing example programs with the aforementioned commands.

TECHNOLOGY MANAGEMENT (IE7XX)

Introduction to Technology Management and the Importance of Technology, Innovation Management, Technology Intelligence and Forecasting, Technology Life Cycle, Types and Models of Innovation Processes, Technology Strategy Development, Technology Transfer Acquisition, Industry 4.0, Blockchain Technology and Cryptocurrencies, NFTs.

Textbook

- Moehrle, M. G., Isenmann, R., & Phaal, R. (2013). Technology roadmapping for strategy and innovation. Charting Route to Success. Berlin, Springer.

- Şenkardeş, Ç. G. (2022). Blockchain Technology and NFTs. CERES PUBLICATIONS.

INTRODUCTION TO DATA MINING (IE7XX)

As an introduction to data mining, this course covers basic Data Preprocessing, Association Rules, Classification and Bunching algorithms and their applications. The last part of the course is devoted to advanced topics in data mining such as Intrusion Detection and Text/Web Mining.

Textbook

- Data Mining, Concepts and Techniques 2nd Edition Jiawei Han and Micheline Kamber ISBN: 978-1-55860-901-3

SEMESTER 8 MANDATORY COURSES

GRADUATION PROJECT-II (IE801)

Each student will carry out a project study on topics related to the industrial engineering department, present it in a report form to their advisor faculty member, and defend it.

APPLIED ENGINEERING TRAINING (IE802)

Applied Engineering Training is a hands-on internship program designed to provide industrial engineering students with practical experience. This course allows students to apply theoretical knowledge in a professional setting, working under the supervision of experienced engineers. Participants will engage in real-world projects, develop essential technical and soft skills, and gain exposure to industry standards and practices. The training aims to enhance students' problem-solving abilities, communication skills, and professional behavior, preparing them for successful industrial engineering careers.

SEMESTER 8 ELECTIVE COURSES

ACADEMIC RESEARCH AND PRESENTATION TECHNIQUES (IE8XX)

Students taking this course will be able to have knowledge about the basic steps of scientific research, data collection, data interpretation, article, report and undergraduate / graduate thesis writing rules, research and publication ethics. In addition, by taking this course, students will have general knowledge about how to search national and international databases, the ability to organize the results of scientific research within the framework of academic writing rules, understanding the types of referencing and presentation techniques.